



AQ5.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Rank of a matrix A is equal to

☐

the row rank of the matrix A .

☐

the column rank of the matrix A .

☐

the nullity of the matrix A .

☐

the number of dependent variables in the system of equations $Ax = 0$.

☐

the number of independent variables in the system of equations $Ax = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

the row rank of the matrix A .

the column rank of the matrix A .

the number of dependent variables in the system of equations $Ax = 0$.

Let nullity of the matrix $A_{3 \times 5}$ be 2. Find the rank of the matrix $A_{3 \times 5}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

If A is an $m \times n$ matrix, then which of the following statements is true?

☐

$\text{rank}(A) + \text{nullity}(A) = m$

☐

$\text{rank}(A) + \text{nullity}(A) = n$

☐

$\text{rank}(A) + \text{nullity}(A) = mn$

☐


$$\text{rank}(A) + \text{nullity}(A) = \max\{m, n\} \text{rank}(A) + \text{nullity}(A) = \max\{m, n\}$$

☐

$$\text{rank}(A) + \text{nullity}(A) = \min\{m, n\} \text{rank}(A) + \text{nullity}(A) = \min\{m, n\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\text{rank}(A) + \text{nullity}(A) = \text{rank}(A) + \text{nullity}(A) = n$$

1 point

Which of the following is true for a homogeneous system of linear equations $Ax = 0$?

- ☐ It may have no solution.
- ☐ It can have infinitely many solutions.
- ☐ It can have a unique solution.
- ☐ Whenever it has a non-trivial (non zero) solution, it must have infinitely many solutions

No, the answer is incorrect.

Score: 0

Accepted Answers:

It can have infinitely many solutions.

It can have a unique solution.

Whenever it has a non-trivial (non zero) solution, it must have infinitely many solutions

1 point

Which of the following options are correct for a square matrix A of order $n \times n$, where n is any natural number?

☐

If the determinant is non-zero, then the nullity of A must be 0.

☐

If the determinant is non-zero, then the nullity of A may be non-zero.

☐

If the nullity of A is non-zero, then the determinant of A must be 0.

☐

If the nullity of A is non-zero, then the determinant of A may be non-zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the determinant is non-zero, then the nullity of A must be 0.

If the nullity of A is non-zero, then the determinant of A must be 0.

1 point

Choose the set of correct statements.

☐

If nullity of a 3×3 matrix is c for some natural number c , $0 \leq c \leq 3$, then the nullity of $-A$ will also be c .

☐

$$\text{nullity}(A + B) = \text{nullity}(A) + \text{nullity}(B)$$

☐

Nullity of the zero matrix of order $n \times n$, is n .

☐

Nullity of the zero matrix of order $n \times n$, is 0.

☐

There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity of $A + B$ is n .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If nullity of a $3 \times 3 \times 3$ matrix is c for some natural number c , $0 \leq c \leq 3$, then the nullity of $-A$ will also be c .

Nullity of the zero matrix of order $n \times n$, is n .

There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity of $A + B$ is n .

The rank of an $n \times n$ matrix all of whose entries are equal to 1 is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

If A is a 3×4 matrix, then which of the following options are true?

☐ rank(A) must be less than or equal to 3.

☐ nullity(A) must be greater than or equal to 1.

☐

If A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then nullity(A) = 2.

☐

If A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then nullity(A) = 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

rank(A) must be less than or equal to 3.

nullity(A) must be greater than or equal to 1.

If A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then nullity(A) = 2.

Level 2

1 point

Let $Ax = 0$ be a homogeneous system of linear equations which has infinitely many solutions, where A is an $m \times n$ matrix (where, $m > 1, n > 1$). Which of the following statements are possible?

☐

rank(A) = m and $m < n$.

☐

rank(A) = m and $m > n$.

☐

rank(A) = m and $m = n$.

☐

nullity(A) = n .

☐

nullity(A) $\neq 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

rank(A) = m and $m < n$.

$$\text{nullity}(A) = n - \text{rank}(A) = n.$$

$$\text{nullity}(A) \neq 0 \Rightarrow \text{rank}(A) = 0.$$

Consider the coefficient matrix A of the following system of linear equations to answer questions 11 and 12:

$$3x_1 + 2x_2 + x_3 = 0 \quad 3x_1 + 2x_2 + x_3 = 0$$

$$x_1 + x_3 = 0 \quad x_1 + x_3 = 0$$

1 point

Which one of the following vector spaces represents the null space of A appropriately?

☐

$$\{(-t, t, t) \mid t \in \mathbb{R}\} \{(-t, t, t) \mid t \in \mathbb{R}\}$$

☐

$$\{(t_1, t_2, \frac{t_2 - t_1}{2}) \mid t_1, t_2 \in \mathbb{R}\} \{(t_1, t_2, 2t_2 - t_1) \mid t_1, t_2 \in \mathbb{R}\}$$

☐

$$\{(t, -t, t) \mid t \in \mathbb{R}\} \{(t, -t, t) \mid t \in \mathbb{R}\}$$

☐

$$\{(t_1, t_2, \frac{t_1 + t_2}{2}) \mid t_1, t_2 \in \mathbb{R}\} \{(t_1, t_2, 2t_1 + t_2) \mid t_1, t_2 \in \mathbb{R}\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(-t, t, t) \mid t \in \mathbb{R}\} \{(-t, t, t) \mid t \in \mathbb{R}\}$$

1 point

What will be the rank of A and nullity of A ?

☐

$$\text{rank}(A) = 3, \text{nullity}(A) = 2 \text{rank}(A) = 3, \text{nullity}(A) = 2$$

☐

$$\text{rank}(A) = 2, \text{nullity}(A) = 1 \text{rank}(A) = 2, \text{nullity}(A) = 1$$

☐

$$\text{rank}(A) = 1, \text{nullity}(A) = 2 \text{rank}(A) = 1, \text{nullity}(A) = 2$$

☐

$$\text{rank}(A) = 2, \text{nullity}(A) = 0 \text{rank}(A) = 2, \text{nullity}(A) = 0$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\text{rank}(A) = 2, \text{nullity}(A) = 1 \text{rank}(A) = 2, \text{nullity}(A) = 1$$

Let $Ax = 0$ be a homogeneous system of linear equations which has a unique solution, where $A \in \mathbb{R}^{n \times n}$. What is the nullity of A ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Suppose $x_1 \neq 0$ solves $Ax = 0$, where $A \in \mathbb{R}^{2 \times 4}$. What is the minimum number of elements in a linearly independent subset of null space of A that also spans the set of solutions of $Ax = 0$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Consider the following sets:

$$S_1 = \{(1, 2, 3), (4, 5, 6), (8, 9, 10)\} S1 = \{(1, 2, 3), (4, 5, 6), (8, 9, 10)\}$$

$$S_2 = \{(1, 2, 1), (2, 4, 2), (1, 2, 5)\} S2 = \{(1, 2, 1), (2, 4, 2), (1, 2, 5)\}$$

$$S_3 = \{(1, 2, 3), (4, 5, 6), (7, 8, 10)\} S3 = \{(1, 2, 3), (4, 5, 6), (7, 8, 10)\}$$

$$S_4 = \{(1, 2), (4, 5)\} S4 = \{(1, 2), (4, 5)\}$$

$$S_5 = \{(1, 2), (4, 5), (3, 8)\} S5 = \{(1, 2), (4, 5), (3, 8)\}$$

Choose the set of correct options

☐

S_1 $S1$ is a basis of R^3 $R3$.

☐

S_2 $S2$ is a basis of R^3 $R3$.

☐

S_3 $S3$ is a basis of R^3 $R3$.

☐

S_4 $S4$ is a basis of R^2 $R2$.

☐

S_5 $S5$ is a basis of R^2 $R2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S_3 $S3$ is a basis of R^3 $R3$.

S_4 $S4$ is a basis of R^2 $R2$.

Check Answers